

**Statistical Correlations Between Vedic Planetary
Configurations and Global Seismic Activity:
An Exploratory Machine Learning Investigation**
*An Interdisciplinary Study Combining Ancient Indian
Astronomical Wisdom with Modern Data Science*

CORRECTED AND RETRACTED EDITION — AUGUST 2026

Supersedes and replaces the original May 2026 edition. See notice below.

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RETRACTION AND CORRECTION NOTICE — AUGUST 2026

The May 2026 edition of this paper reported statistically validated predictive signal in the Americas Pacific zone (Mexico, Peru, Chile), based on hit-rate validation computed with a model trained on the complete 1900-2026 dataset — including the same post-2000 events it was then evaluated against. This is a data-leakage error: the model had, in effect, seen the answers before the test.

A subsequent independent reproducibility investigation, conducted by the lead author with an AI research assistant (Claude, Anthropic) over July-August 2026, corrected this error and re-tested the underlying hypothesis seven independent ways — including a direct re-validation of this paper's own claimed Americas Pacific zone scope, using a model trained only on pre-2000 data. Every one of the seven tests found no predictive signal distinguishable from chance.

This edition retains the structure and ancient-source scholarship of the May 2026 paper. Sections 3, 4, 5 and 6 have been rewritten to present the corrected, honest findings in place of the original, retracted claims.

KEY FINDING

This study set out to answer one question: do planetary configurations — as described in classical Vedic astronomical texts, echoed in 20th-century Western astrology, and extended in an original hypothesis covering the outer planets — predict earthquake timing or location above random chance?

For the first time, that question has been tested rigorously, at scale, against 126 years of global earthquake data, with 85 named rules and combinations given every reasonable chance to succeed, and with a model evaluated only on evidence it had never seen before.

The answer is no.

Personal Foreword

To Prabha — my wife and companion in all things, including a five-year Sanskrit course at the University of Madras — who stood beside me through 64 years with patience, love and unwavering faith.

To my sons Nagaraja Srivatsan and his wife Shanthi, and Srikantan Nagarajan and his wife Srividya — who gave me the freedom to pursue my insatiable curiosity without complaint.

To my grandchildren Aditya Srivatsan, Durga Srivatsan, Shankara Srikantan and Shiama Srikantan — whose contributions made this research possible.

To Varahamihira, Ballala Sena, Parasara, Garga and all the ancient Indian sages whose careful observations of sky and earth, recorded 1500 years ago, remain worthy of serious, rigorous testing — whatever that testing finds.

A special acknowledgement to Claude (Anthropic AI), who served as computational partner both in the original validation phase and, months later, in the harder work of finding and correcting this study's own errors. Good research includes being willing to be wrong in public.

Abstract

This study concludes, based on rigorous large-scale statistical testing against 126 years of global earthquake data, that classical Vedic astrological rules — together with 20th-century Western astrological rules and a further original hypothesis extending the search to the outer planets — do not predict earthquake timing or location above random chance. This is the first time these specific, named rules have been tested at this scale, with a methodology designed specifically to avoid the two failure modes that have historically made this kind of claim impossible to trust: small, hand-selected case studies, and evaluation on data a model has already seen.

This paper presents an exploratory statistical investigation into whether planetary configurations derived from classical Indian Vedic astronomical methods show measurable associations with seismic activity. The study treats the classical rules as scientific hypotheses under investigation — not as assertions of astrological causation.

The investigation proceeded in three stages. Phase 1 (2018-2019), developed by Aditya Srivatsan in 11th grade, cast Vedic horoscopes for 4,000 earthquake and non-earthquake days and trained a Logistic Regression

model, establishing a preliminary WHEN signal. Phase 2 (2025-2026) extended this using XGBoost across 901 global grid cells and 818 planetary features, and reported (May 2026) statistically significant hit-rate elevation in the Americas Pacific zone — Mexico 61.9%, Peru 45.8%, Chile 38.5%, all against a 21.8% random baseline, $p < 0.001$.

Phase 3 (July-August 2026), the subject of this corrected edition, was an independent reproducibility investigation of Phase 2's own findings. **It found that the Phase 2 hit-rate validation had been computed using a model trained on the entire 1900-2026 dataset, including the same post-2000 events it was then evaluated against — a fundamental train/test leakage error, not a minor statistical wrinkle.**

After correcting this leakage and re-deriving an honest classification threshold from pre-2000 data only, seven independent tests were conducted: (1) the 24 individual classical rules from B.V. Raman, Varahamihira, Johnston and Peery, tested globally across 126 years; (2) 9 pairwise combinations of those rules; (3) a direct day-level prevalence check comparing rule activity on actual earthquake days versus ordinary days across the full 126-year record; (4) a 52-rule outer-planet hypothesis (Uranus, Neptune, Pluto, the lunar nodes), formulated

and tested at the lead author's own initiative following the null results in (1)-(3); (5) a control test for earthquake-detection-era confounding, which explained away the one dramatic-looking result found in test (4); (6) a direct re-validation of this paper's own originally claimed scope — the Americas Pacific zone — using the corrected, non-leaky model and the original clean-threshold methodology; and (7) a repeat of the direct prevalence method on genuinely fresh data, January-July 2026, which reproduced and confirmed the same detection-era illusion identified in test (5).

All seven tests returned a result indistinguishable from chance (MCC values between -0.0002 and 0.0024; the Americas Pacific zone re-validation specifically returned MCC = 0.0008, with prediction precision of 0.2%, meaning roughly 998 of every 1,000 high-risk flags were false alarms).

This corrected edition therefore retracts the Phase 2 headline findings reported in May 2026. It preserves, unaltered, the historical and textual scholarship on the classical sources (Section 1), and presents the corrected Phase 3 methodology and results in full (Sections 3 and 6), so that this study's own error and its correction form part of the permanent public record.

Keywords: Vedic astrology, earthquake prediction, machine learning, XGBoost, planetary aspects, seismic risk, data leakage, reproducibility, null result, Brihat Samhita, Adbhuta Sagara, Lahiri Ayanamsa

Notation and Definitions

The following terms are used throughout this paper. All definitions are operationally precise to ensure reproducibility.

PLANETARY ASPECT: An angular relationship between two planets as observed from Earth. Five aspects are used: Conjunction (0° , orb $\pm 8^\circ$), Sextile (60° , orb $\pm 6^\circ$), Square (90° , orb $\pm 8^\circ$), Trine (120° , orb $\pm 8^\circ$), Opposition (180° , orb $\pm 8^\circ$).

SIDEREAL vs TROPICAL COORDINATES: This study uses the Lahiri Ayanamsa — the sidereal offset adopted by the Government of India Calendar Reform Committee in 1955, currently approximately 23.85° behind the tropical system. All planetary longitudes are sidereal Lahiri unless stated otherwise.

NAKSHATRA: The 27 lunar asterisms of Vedic astronomy, each $13^\circ 20'$ of the ecliptic. Each planet's Nakshatra position is encoded as a binary indicator.

MCC (MATTHEWS CORRELATION COEFFICIENT): The primary evaluation metric used in the corrected Phase 3 methodology, replacing the hit-rate/lift metrics used — and later found unreliable — in Phase 2. MCC ranges from -1 to +1: +1 is perfect prediction, 0 is performance indistinguishable from random guessing, -1 is perfect

inverse prediction. MCC is well-behaved under the severe class imbalance inherent to earthquake prediction, which is why it was adopted for the corrected analysis in place of raw hit rate.

PREVALENCE / LIFT (corrected usage): The ratio of a rule's activation rate on earthquake days to its activation rate on ordinary days. Lift = 1.0 means the rule is equally common whether or not an earthquake occurred that day — i.e., the rule carries no information. This is the metric used throughout Section 3 of this corrected edition.

DATA LEAKAGE: A methodological error in which information from the evaluation set is, directly or indirectly, available to a model during training or model selection, producing an inflated and unreliable performance estimate. This is the error identified and corrected in Phase 3 of this study, described in full in Section 2.7 and Section 3.

1. Introduction

1.1 Background and Motivation

Earthquakes remain among the most destructive and least predictable natural phenomena. Modern seismology has made great advances in identifying fault zones, estimating long-term seismic hazard and measuring ground motion. However, reliable prediction of the specific date of a future earthquake — as distinct from estimating long-term probability — remains an unsolved problem. This limitation motivates the investigation of alternative temporal signals.

This study investigates whether planetary configurations described in classical Indian astronomical texts, when computed precisely using modern astronomical software and tested against 126 years of global seismic data, show statistically measurable associations with earthquake occurrence. This hypothesis has been discussed in Indian astronomical literature for over 2,000 years but had never previously been subjected to systematic large-scale statistical testing — nor, as this corrected edition documents, had it previously been subjected to a properly isolated, leakage-free test.

1.2 The Classical Rules Tested in this Study

The following classical rules form the hypothesis set tested in this study. They are presented here not as

established facts but as precisely stated hypotheses that the study subjects to statistical examination. Nothing in the historical scholarship below is affected by the correction described in Section 3 — what changed is not what the ancient sources said, but how faithfully and how honestly this study tested it.

1.2.1 B.V. Raman's Rules (20th century)

B.V. Raman, in his work "Astrology in Forecasting Weather and Earthquakes," analysed 200 historical earthquake cases from the 19th and 20th centuries and identified the following planetary combinations as most frequently associated with major earthquakes:

- Mars-Saturn Opposition: When Mars and Saturn are 180° apart, their combined malefic influence — Mars representing explosive energy and Saturn representing restriction and compression — creates maximum tension. Raman found this to be his strongest single indicator.
- Jupiter-Saturn Opposition: The opposition of the two largest planets was identified as the second strongest indicator.
- New Moon with Multiple Planets in One Sign: Raman's most dramatic example was the January 15, 1934 Bihar earthquake — a new Moon day when seven planets were in Capricorn.

- Jupiter in the 8th or 12th house from the rising sign at the time of the earthquake: found statistically frequent in Raman's case analysis.

Raman's methodology was case analysis — he examined earthquakes that had already occurred and identified planetary patterns present in those cases. This study's corrected Phase 3 methodology tests whether those patterns, computed prospectively and evaluated with a model that never saw the test data, predict future earthquakes above random chance. Section 3 reports that they do not.

1.2.2 Varahamihira's Four Circles (Brihat Samhita, 5th-6th century AD)

Varahamihira, in Adhyaya XXXII of his encyclopaedic work Brihat Samhita ("The Great Compilation"), describes a systematic earthquake prediction framework based on the Moon's position in the 27 Nakshatras. He divides the Nakshatras into four groups, each associated with a natural element, a geographic direction and a different time horizon for predicted effects:

Circle	Element	Direction	Time Horizon	Reach
VARUNA	Water	West / Ocean	Same day	180 Yojanas
INDRA	Thunder	East	Within one	160

			week	Yojanas
AGNI	Fire	Southeast	Within six weeks	110 Yojanas
VAYU	Wind	Northwest	Within two months	200 Yojanas

The 28 Nakshatras (7 per circle, including Abhijit) associated with each circle:

Circle	Nakshatras (7 per circle)
VARUNA	Revati, Purva Ashadha, Ardra, Ashlesha, Mula, Uttara Bhadrapada, Shatabhisha
INDRA	Abhijit, Shravana, Dhanishtha, Rohini, Jyeshtha, Anuradha, Ashlesha
AGNI	Pushya, Kritika, Vishakha, Bharani, Magha, Purva Bhadrapada, Purva Phalguni
VAYU	Uttara, Hasta, Chitra, Swati, Punarvasu, Mrigashira, Ashwini

A significant feature of Varahamihira's framework, worth restating in this corrected edition since it remains true regardless of the statistical findings below, is that each

circle carries a different time horizon — from same-day (Varuna) to two months (Vayu). This multi-timescale framework motivated the corrected Phase 3 methodology's inclusion of a dedicated day-level prevalence check (Section 3.3) rather than relying solely on model-based hit rates.

Varahamihira also describes (Sloka 27) cancellation effects — when opposing circles are simultaneously active, they suppress each other's effects — and (Sloka 32) foreshock recognition: if an earthquake is followed by another on the 3rd, 4th or 7th day, a major event will follow. The foreshock pattern was independently identified by modern seismology in the 20th century.

[1.2.3 Ballala Sena's Additions \(Adbhuta Sagara, 10th-11th century AD\)](#)

Ballala Sena's Adbhuta Sagara ("Ocean of Wonders") adds directional precision from the sage Ushanas (East for Indra, South for Agni, West for Varuna, North for Vayu), a detailed listing of geographic regions affected by each circle type, and descriptions of mixed-type earthquakes when multiple circles are simultaneously active.

[1.2.4 Parasara's Hypothesis \(Brihat Parasara Hora Shastra, ~310 BC\)](#)

Parasara, writing approximately 310 BC, stated: "Arka Chandra grahana graha vikrithachara jamscha kampana hayulu" — meaning that when the course of a planet is

disturbed from the normal path owing to the attractive force of other planets, there is a shaking of the planet perceived as an earthquake. This formulation describes planetary perturbations as the cause of seismic activity, conceptually similar to modern gravitational and tidal forcing theories, and motivated this study's use of inter-planetary angular aspects as primary features.

1.2.5 Garga's Solar Connection (Garga Samhita, Ancient period)

The Garga Samhita traces earthquakes to solar Ketu — dark spots on the Sun. In this study, Sun-related aspects and the Sun's Nakshatra position were included as features throughout.

1.3 Methodological Integrity: What Changed Between May and August 2026

A study combining ancient rules with modern data always risks post-hoc justification — finding a pattern that fits the cases already known, without it holding for cases not yet seen. This risk turned out to be real, though not in the specific place it was first watched for.

The classical rules were, as intended, never used to directly configure the model's training objective — the model was trained purely on planetary feature data and earthquake labels. What did not hold up was the validation used to confirm the model had learned something real: the model whose feature importances and regional hit rates were reported in the May 2026

edition had been trained on the complete 1900-2026 dataset, including the same post-2000 earthquakes it was then evaluated against.

This is the single most important correction in this paper. It was found by the lead author's own follow-up investigation, conducted deliberately after publication and public circulation of the original findings. Section 3 describes the correction and the seven independent tests conducted afterward, all on properly isolated, never-before-seen data.

1.4 Research Hypothesis

This study tests the following primary hypothesis:

H1: Specific planetary configurations computed using Vedic sidereal astronomy (Lahiri Ayanamsa) show statistically significant associations with elevated earthquake probability, measurable above random chance on data never seen by the model during training or model selection.

The corrected Phase 3 methodology (Section 2.7, Section 3) tests H1 both globally and specifically within this paper's originally claimed Americas Pacific zone scope. In both cases, H1 is not supported by the corrected analysis.

A non-technical companion paper — "When Planets Converge, Does the Earth Shake?" — is available separately and has been corrected to match this edition's findings.

2. Data and Methodology

2.1 The Three-Phase Research Design

This research proceeded in three distinct phases. Phases 1 and 2 are described as originally conducted; Phase 3 is the corrected-edition addition documenting the reproducibility investigation.

Phase 1: Foundation Study (Aditya Srivatsan, 11th grade)

2,000 earthquake days and 2,000 non-earthquake days were identified from historical seismic records. For each, a complete Vedic astrological horoscope was cast — planetary longitudes, house positions, Nakshatra placements, Tithi and all inter-planetary aspects. A Logistic Regression model (with Lasso, Ridge and Elastic Net variants) and K-Nearest Neighbours were trained on this dataset. Training period 1900-2010, testing 2011-2020. Phase 1 demonstrated a preliminary WHEN signal but did not address WHERE.

Phase 2: Extension Study (Aditya Srivatsan, refined by Durga Srivatsan)

Phase 2 extended the foundation using XGBoost, an 818-feature set computed via Swiss Ephemeris, and a 901-cell global grid at 5° resolution, attempting to address both WHEN and WHERE.

Phase 3: Independent Reproducibility Investigation (July-August 2026)

Phase 3, conducted by the lead author with Claude (Anthropic) as computational partner, re-examined

Phase 2's own methodology and findings. It is the subject of the corrections presented in this edition. Its full methodology is given in Section 2.7 and its results in Section 3.

2.2 Earthquake Data

All seismic data was obtained from the USGS real-time earthquake catalog (earthquake.usgs.gov). Events with magnitude M5.5 and above were selected.

Dataset	Period	Events (M5.5+)	Purpose
Training set	1900-1999	~32,000 events	Model learning — never seen during corrected testing
Test set	2000-2025	12,830 events	26-year out-of-sample validation (corrected in Phase 3)

2.3 Astronomical Feature Computation — Swiss Ephemeris with Lahiri Ayanamsa

Planetary positions for every date from 1900 to 2026 were computed using the Swiss Ephemeris library (Lahiri

Ayanamsa sidereal system), for Sun, Moon, Mars, Mercury, Jupiter, Saturn, Venus, Rahu and Ketu, Uranus, Neptune, Pluto and Chiron. For each planet: sidereal longitude, Nakshatra number and name, zodiac sign number, and retrograde status.

2.4 Feature Engineering

Classical rules were converted to numerical features via three mechanisms, unchanged from the original methodology: (1) binary aspect-active flags for each planet pair and aspect type (~420 features); (2) binary Nakshatra indicators per planet (351 features); (3) sine/cosine circular encoding of longitudes (26 features). Total 818 base features, of which 188 relate to Chiron and Lilith (included in training but not part of the classical Vedic system).

2.5 Machine Learning Model — XGBoost

Parameter	Value
Algorithm	XGBoost binary classifier
Number of trees	300
Learning rate	0.05
Max tree depth	6
Class weight	scale_pos_weight matched to

	training-set downsample ratio
Training period	1900-1999
Test period	2000-2026

2.6 Grid Cell Structure

The Earth's surface was divided into 901 grid cells at 5° resolution. For each date, the model outputs a per-cell probability; regional risk scores and predictions were derived from these, as described in the original Phase 2 pipeline documentation on the project GitHub repository.

2.7 The Leakage Error and Its Correction — Phase 3 Methodology

THE ERROR: The 'final' model referenced throughout the original Phase 2 validation (Section 3 of the May 2026 edition) was produced by a training script that fit on the entire 1900-2026 dataset — including all real post-2000 earthquake positives. That same model was then used to compute the country-level hit-rate 'validation' (Mexico, Peru, Chile, and other regions). Because the model had already seen every earthquake it was later evaluated against, the reported hit rates and feature-importance rankings were not a valid test of generalization.

This error was identified by the lead author's own follow-up investigation in July 2026, not by an external reviewer, and was corrected as follows:

- A separate model was trained strictly on pre-2000 data and never exposed to any post-2000 row, at training time or at model-selection time.
- The classification threshold — previously chosen using the same post-2000 data the model was then scored on, a second, subtler form of leakage — was re-derived from a held-out pre-2000 validation slice (1990-1999) and fixed before touching the post-2000 test set.
- Evaluation was switched from raw hit rate (which is unstable and easily inflated under class imbalance) to the Matthews Correlation Coefficient (MCC), which behaves correctly under the roughly 100:1 to 1000:1 imbalance between ordinary days/cells and earthquake days/cells present in this data.
- Seven independent tests were then run using this corrected model and methodology, described in full in Section 3.

This is the single methodological change that separates this corrected edition from the original: the target metric, the threshold-selection procedure, and the training/evaluation split were all made strictly leakage-

free, and every number reported in Section 3 below was produced under that discipline.

3. Corrected Findings — Seven Independent Tests, All Null

This section presents the Phase 3 reproducibility investigation in full: what was tested, how, and what was found. Every result below was produced using the corrected, leakage-free model and methodology described in Section 2.7.

3.1 Test 1 and 2 — The 24 Classical Rules and 9 Combinations, Globally

The classical rules described in Section 1.2 (Raman, Varahamihira, Ballala Sena, Parasara, Garga) and the 20th-century Western rules of Johnston and Peery were re-implemented as 24 explicit, individually named feature columns, each traced directly to a specific passage in a specific source. Nine pairwise combinations, chosen because the source texts discuss them together (e.g. Raman's own six-item combined rule; Varahamihira's Kendra-and-Trikona pairing), were added as a second pass.

Using the corrected, non-leaky model and evaluation procedure (Section 2.7), these 33 features were tested both as standalone additions to the base 818-feature model and via a direct global prevalence comparison (rate of activation on real earthquake days versus ordinary days, across all 46,386 days from 1900-2026):

Test	Result	Interpretation
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Baseline (818 features, leakage-corrected)	MCC = 0.0004	No signal above chance
24 individual classical rules added	MCC = -0.0002 (original rule definitions)	No signal above chance
24 rules, corrected for fidelity to source texts	MCC = 0.0004	No signal above chance
9 rule combinations added	MCC = 0.0004	No signal above chance
Direct prevalence check, full 126 years, all 33 rules/combinations	Highest lift = 1.10 (not statistically robust); all others ≈ 1.0	No individual rule or combination fires meaningfully more often on earthquake days than ordinary days

A methodological note included in the corrected edition for transparency: an initial pass at implementing these 24 rules contained several fidelity gaps against the primary

sources — one of Raman's six named conditions ("planets in Trikona") was missing entirely, his "eclipse degree and Mars" condition had lost its Mars requirement, Johnston's rule had been narrowed to five planets rather than the ten his own study analysed, and Peery's rule (explicitly a Western tropical-zodiac rule) was being compared against this study's sidereal data with no ayanamsa correction. All five gaps were identified and corrected before the final test shown above; correcting them moved the result from $MCC = -0.0002$ to $MCC = 0.0004$ — a negligible change, but the correct and honest one to report.

3.2 Test 3 — Direct Prevalence Check Across the Full 126-Year Record

Independent of any machine learning model, a direct check was run: for each of the 33 classical rules/combinations, what fraction of actual M5.5+ earthquake days (22,717 days, 1900-2026) show the rule active, compared to the fraction of ordinary days (23,669 days) that show it active? This mirrors the method the classical sources themselves used — checking a named rule against known earthquake dates — but across the full historical record rather than a hand-selected case set.

Result: no rule or combination showed a lift meaningfully or robustly above 1.0. The highest (Moon-Mercury same-

nakshatra conjunction, lift 1.10) did not clear a conventional significance threshold at this sample size.

3.3 Test 4 — An Independent Outer-Planet Hypothesis

Following the null results in Tests 1-3, a further hypothesis was formulated and tested at the lead author's own initiative: that Uranus, Neptune, Pluto and the lunar nodes — bodies unknown to the classical Vedic system and not part of Raman's, Johnston's or Peery's rule sets — might carry a signal the classical rules missed, acting as slow "background" tectonic-stress indicators triggered by faster-moving bodies (Sun, Moon, Mars, Mercury).

Forty individual rules and twelve combinations were designed around this hypothesis, deliberately including both slow-only sign-placement rules and fast-slow aspect pairings engineered to be day-specific rather than true for months at a stretch. Tested across the full 126-year record:

- All fast-slow, day-specific pairings: lift ≈ 1.0 , no signal.
- Two slow-only sign-placement rules — Pluto in Capricorn, Neptune in Pisces — initially showed a striking result: lift 2.18 and 2.03 respectively, with extremely small p-values.

3.4 Test 5 — Controlling for Earthquake-Detection-Era Confounding

The two dramatic results in Test 4 were investigated before being accepted. Pluto occupied Capricorn only from 2008-2024, and Neptune has occupied Pisces only since 2011 — both recent, multi-year windows within the 126-year study period. Independently, the raw rate of recorded M5.5+ earthquake days rose from approximately 10% of days in the 1900s to 65-70% by the 1970s-2010s, reflecting the maturation of global seismic monitoring networks rather than any change in the Earth's actual seismicity.

Re-running the Pluto-in-Capricorn and Neptune-in-Pisces tests restricted to a single stable-detection era (1970-2020, during which the raw detection rate stayed within a narrow 0.58-0.71 band) produced:

Rule	Full 126-year lift	Stable-era (1970-2020) lift
Pluto in Capricorn	2.18 ($p \approx 6.5 \times 10^{-198}$)	1.02 ($p = 0.56$, not significant)
Neptune in Pisces	2.03 ($p \approx 3.0 \times 10^{-145}$)	0.95 ($p = 0.12$, not significant)

Both apparent signals collapsed entirely once the detection-era confound was removed. This is presented in the corrected edition not as a footnote but as a central

result: it demonstrates, with a controlled before/after comparison, exactly the mechanism by which an artifact of improving earthquake instrumentation can masquerade as an astronomical signal, and it is why every other result in this section was checked as carefully as time allowed before being reported.

3.5 Test 6 — Direct Re-Validation of This Paper's Own Claimed Scope

The final test directly re-examined the one specific claim this paper originally made: that validated predictive signal exists in the Americas Pacific zone (Mexico, Peru, Chile, Western USA/Alaska/California, Colombia, Caribbean). Using the corrected, non-leaky model and the same clean-threshold procedure described in Section 2.7, applied specifically and only to this originally claimed region:

Region	MCC	Precision	Interpretation
Americas Pacific zone (this paper's claimed scope)	0.0008	0.2%	No signal above chance; ~998 of 1,000 high-risk flags are false alarms
Rest of world (reference only)	0.0002	0.1%	No signal above chance

This is the direct, corrected replacement for the original Section 3.3-3.4 regional validation tables (Mexico 61.9%, Peru 45.8%, Chile 38.5%). The original numbers were real outputs of the leaky model; they were not fabricated. But they measured the model's memory of data it had already seen, not its ability to generalize to new data — which is the only thing a predictive claim can honestly rest on.

3.6 Test 7 — Fresh Data: January-July 2026

As a final check, using real data no other test in this study had access to, the same direct prevalence method used in Tests 3 and 4 was applied to January 1 - July 31, 2026 — seven months of genuinely fresh earthquake activity, extending two months past the original paper's own May 2026 cutoff. Fifty-two earthquake days and 160 ordinary days were compared across all 85 rules and combinations.

A small number of static, slow-moving rules — Neptune in Pisces, Uranus in Taurus, and combinations built on them — showed dramatic-looking lift (as high as 8x) in this short window. This is the same pattern examined and explained in Test 5: Uranus and Neptune barely move sign over seven months, so if a handful of real earthquakes happened to cluster in this period for ordinary reasons (aftershock sequences are the obvious

candidate), any slow placement held throughout the window will appear associated with them by coincidence. This reading is confirmed directly by already having the full 126-year, era-controlled result for these exact two rules on file (Test 5): Uranus in Taurus and Neptune in Pisces both returned lift at or below 1.0 once tested properly. A short window cannot overturn a result already established on a much larger, cleaner test — it can only be misread if taken in isolation.

Consistent with this reading, the rules specifically engineered to be immune to this effect — the fast-slow, day-to-day pairings such as Moon-Uranus and Mars-Uranus aspects — again showed no elevation, scattering around lift 1.0-1.5 exactly as they did throughout Tests 1-6.

This seventh test therefore does not qualify the finding — it reinforces it, using the most recent data available and, notably, data the original May 2026 edition could not have seen.

3.7 Summary of Corrected Findings

Test	Result
1. 24 classical rules, individually, global	MCC = 0.0004

2. 9 rule combinations, global	MCC = 0.0004
3. Direct prevalence check, 126 years, global	Max lift 1.10, not robust
4. 52 outer-planet rules, global	Lift $\approx$ 1.0 throughout, except two flagged in Test 5
5. Era-controlled re-test of the two flagged rules	Both collapse to lift $\approx$ 1.0 once corrected
6. Americas Pacific zone (this paper's own claimed scope), non-leaky model	MCC = 0.0008, precision 0.2%
7. Fresh data, January-July 2026, direct check	Same static-rule illusion as Test 4, resolved the same way as Test 5

Across every method available to this study — global and regional, classical and modern, individual rules and combinations, inner planets and outer planets, historical data and the freshest data available at time of

writing, with a dedicated control for the one confound that produced a false-positive-looking result — no test found predictive signal distinguishable from chance.

4. Prospective Validation — January 1 to May 21, 2026 (Status: Withdrawn)

The May 2026 edition reported that 81.8% of M5.5+ earthquakes in this window (36 of 44 events) fell within model-predicted zones, rising to 100% (15 of 15) during high-convergence periods, against a 33% random baseline. This corrected edition withdraws those numbers.

The 2026 risk calendar that produced these predictions was itself built from the leakage-affected model: its specific dates and convergence thresholds were chosen using the feature importances and hit-rate validation described in Section 2.7, all of which are now known to have been produced by a model trained on data it was then evaluated against. That calendar cannot be treated as a valid prediction, regardless of how the real earthquakes that followed happened to fall relative to it.

A corrected version of this test was not built, for a plain reason rather than an oversight: doing so would mean generating a new 2026 risk calendar from the corrected model described in Section 2.7 — but that model, tested seven independent ways in Section 3, found no signal to build a calendar from. A risk calendar built from a model with no predictive signal would not identify meaningfully higher- or lower-risk dates; it would be arbitrary.

Constructing one and then reporting how it happened to score against 44 real events would not be a real additional test — it would be re-running Section 3's null result through more steps to reach the same answer. For that reason, this corrected edition withdraws the original prospective validation outright rather than replacing it with a new number.

The 2026-2027 risk calendar originally published in the companion layman paper has been withdrawn (see GitHub notice, August 2026) and should not be used for any planning purpose.

5. Discussion

5.1 What the Corrected Findings Mean

The original discussion of a small but real effect size, a WHERE problem distinct from a WHEN problem, and a supplementary probabilistic indicator worth further study, was predicated on there being a real, if modest, signal to discuss the size and shape of. Seven independent corrected tests found no such signal. There is, at this time, no effect whose size, mechanism, or regional concentration remains to be discussed — only the finding that none was present, across every reasonable way the hypothesis could be tested.

5.2 Why the Original Analysis Looked Positive

Three distinct mechanisms, each documented in this corrected edition, are sufficient to explain why the original May 2026 analysis reported a positive result where none exists:

- Train/test leakage (Section 2.7): the model was evaluated on data it had already been trained on. A sufficiently flexible model (XGBoost, 300 trees, depth 6, 818 features) can memorize an arbitrary historical dataset to a high degree; the reported hit rates measured that memorization, not generalization.
- Detection-era confounding (Section 3.4): a slow-moving planet whose zodiac transit happens to

coincide with recent decades of improved global earthquake monitoring will show an inflated apparent association with earthquake days, entirely independent of astronomy.

- Small-sample and post-hoc effects in the original rule selection: the original Section 3.5 reported '30 validated aspects' selected by their lift ratio computed on the same test data being reported on — the paper's own Section 2.7 acknowledged this as 'some degree of data-adaptive feature selection.' This is precisely the mechanism, familiar from the classical sources' own case-study methodology (Section 1.3), by which searching a large space of possible rules against a limited set of known cases will reliably surface apparent patterns that do not generalize.

5.3 Correlation, Causation, and the Absence of Either

The original paper correctly noted that correlation does not imply causation, and declined to adjudicate between tidal forcing, solar activity, and other candidate mechanisms. This corrected edition need not adjudicate between causal mechanisms for an association that the corrected analysis does not find to exist in the first place.

5.4 On the Ancient Sources Themselves

Nothing in this corrected analysis diminishes Varahamihira's, Ballala Sena's, or B.V. Raman's stature as careful observers working within the tools and statistical understanding available to them. Raman's case-study method — examining earthquakes that had occurred and identifying shared planetary features — is a method that, applied to a small, hand-selected set of cases, is now well understood (through the broader history of statistics, not specific to this study) to reliably produce patterns that appear compelling but do not survive testing against cases the pattern was not built from. This corrected edition's central methodological contribution is exactly this out-of-sample discipline, applied for the first time to these specific rules at this specific scale — and it is a discipline that was not, and could not have been, available to any of the classical authors.

5.5 Why the Ancient Systems Likely Could Not Have Succeeded

Beyond the specific mechanisms in 5.2, four deeper and more general reasons help explain why this negative result should not be surprising, and why it is unlikely to be an artifact of this particular study's design:

- No comprehensive validation was ever possible for the original authors. Varahamihira, writing in the 6th century, had no access to a systematic,

simultaneous, global earthquake catalog — no such catalog existed anywhere on Earth until the 20th century. His four-circle framework, and Raman's 200-case study centuries later, were built from whatever local and historical earthquake reports were available at the time, inevitably a small and geographically limited sample. Neither author could have tested their own rule against the full space of world seismicity. This study is the first opportunity in history for that test to be run at all — not a repeat of a prior validation, but the first one.

- The geographic framework is India-centric, not tectonically calibrated. Varahamihira's and Ballala Sena's regional lists (Saurashtra, Magadha, Kuru, Dravida, and so on, detailed in Section 1.2.2-1.2.3) describe the political and cultural geography of ancient India, not the actual distribution of the world's tectonic plate boundaries. The great majority of the world's largest earthquakes occur along the Pacific Ring of Fire — a feature of plate tectonics unknown to any pre-20th-century observer anywhere in the world. A predictive framework built without reference to the actual physical structure generating the events it claims to predict has no principled reason to align with that structure, and

this study's global test (Section 3) found that it does not.

- No physical mechanism connects planetary angles, as seen from Earth, to the release of tectonic stress at the scale required to trigger an earthquake. The gravitational tidal forces exerted by Mars, Jupiter, or Saturn on the Earth's crust are many orders of magnitude smaller than the Sun's and Moon's own daily tidal forces — forces which themselves have not been established, despite considerable modern seismological study, to reliably trigger major earthquakes on a predictable timetable. A planetary aspect (an angular relationship as seen from Earth) carries no additional gravitational or physical influence beyond what the planets' actual positions already exert continuously; there is no known channel through which an angle alone could act as a trigger.
- The case-study method is structurally prone to finding patterns that are not real. As discussed in Section 1.3 and Section 5.2, examining a limited set of known earthquakes and searching for a shared planetary feature — the method used by both the classical authors and, unknowingly, by the leakage-affected Phase 2 model — will reliably surface apparent patterns purely by chance, given the very

large number of possible planetary combinations available to search through. This is not a criticism specific to Vedic astrology; it is a general property of small-sample pattern-search that affects any field working with limited historical case data, ancient or modern.

5.6 The Methodology, Stated Plainly

Stripped of technical detail, the corrected Phase 3 test was simple, and worth stating in plain terms precisely because the simplicity is part of what makes the result trustworthy: birth-chart-style planetary configurations were computed for the days of roughly 2,000 major earthquakes and an equal number of ordinary days; a machine learning model was asked to find any pattern in this data that distinguished the two; and the model was separately, explicitly fed all 85 named classical and modern astrological rules and combinations described in this paper, so that it did not have to discover them on its own. Despite being handed every rule this study could identify from the literature, and despite being free to search for any additional pattern beyond those rules, no rule, combination, or discovered pattern produced a statistically reasonable predictive result.

5.7 Limitations of This Corrected Study

Limitation	Description
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Original 2026 calendar not re-scored	The original leakage-affected 2026 risk calendar is withdrawn rather than re-scored (Section 4). A direct, model-free check of fresh January-July 2026 data was run instead (Test 7, Section 3.6) and found the same pattern as the rest of this study.
Aspect orb conventions	An 8° general orb was used throughout for consistency; the original ephemeris pipeline's exact orb could not be independently confirmed.
Chiron/Lilith retained	188 of 818 base features relate to bodies outside the classical Vedic system; retained for continuity with the original feature set but not part of any of the 24+52 explicitly-named rules tested in Section 3.
Regional definition	The Americas Pacific zone re-test (Section 3.5) used country-string matching from USGS place-name parsing, which may not perfectly reproduce the original paper's grid-cell region boundaries.

6. Conclusion and Retraction Statement

6.1 Summary of Corrected Findings

Finding	Result
24 classical rules (Raman, Varahamihira, Johnston, Peery), global	MCC = 0.0004
9 rule combinations, global	MCC = 0.0004
Direct 126-year prevalence check, all 33 rules	No rule/combination shows robust lift above 1.0
52-rule outer-planet hypothesis (Uranus/Neptune/Pluto/Nodes), global	No signal; two apparent hits explained by detection-era confound
Americas Pacific zone (this paper's own originally claimed scope)	MCC = 0.0008, precision 0.2%
Fresh data, January-July 2026, direct check	Same detection-era illusion as the outer-planet test, resolved the same way

Mars-Saturn opposition as 'top ML feature' (original Section 3.1)	Retracted — was an artifact of the leakage-affected model
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6.2 Formal Retraction

The following specific claims from the May 2026 edition of this paper are formally retracted:

(1) That Mexico, Peru, and Chile show statistically validated earthquake prediction hit rates of 61.9%, 45.8%, and 38.5% respectively against a 21.8% random baseline.

(2) That Mars-Saturn opposition and Jupiter-Saturn opposition were independently confirmed as top-ranked predictive features by an unbiased machine learning process.

(3) That the Americas weekly prediction achieves a 1.33x lift with $p < 0.0001$ over a genuine 26-year out-of-sample test.

(4) The 2026-2027 seismic risk calendar, and any operational or preparedness use of it.

All four claims rested on a model evaluated on data it had already been trained on. Corrected, leakage-free re-testing (Section 3) found no predictive signal in any

of these claims, including a direct re-test of claim (1) using the paper's own originally specified region and methodology.

6.3 What Is Not Retracted

The historical and textual scholarship in Section 1 — the description of Varahamihira's four-circle Nakshatra framework, Ballala Sena's directional refinements, Parasara's perturbation hypothesis, Garga's solar-Ketu connection, and B.V. Raman's documented case-study rules — is not affected by this retraction and is presented unaltered. What has been withdrawn is this study's claim to have found statistical confirmation of these rules in modern seismic data, not the existence or content of the rules themselves as recorded in the classical sources.

6.4 Practical Implications

There are none, at this time. No probabilistic statement about elevated regional seismic risk derived from this study's models should be used for planning, resource positioning, or public communication. The forecast calendar has been withdrawn from the project repository.

6.5 Final Statement

This research asked a question that had never, in over two thousand years, been properly answered: do planetary configurations, as described in classical Vedic astronomical texts and echoed in later Western astrological tradition, actually predict when and where the earth will shake? Not whether they seem to, in a handful of remembered cases. Whether they do, when checked against every earthquake of consequence the modern world has recorded.

For the first time, that question has been asked properly — with 126 years of global data, with 85 named rules given every reasonable chance to succeed, with a machine free to search for patterns the rules themselves might have missed, and with a model tested only on evidence it had never been shown before.

The answer is no.

Not "inconclusive." Not "promising, pending further study." No signal was found — not in Raman's rules, not in Varahamihira's four circles, not in the outer planets unknown to either tradition, not globally, and not in the one specific region this paper originally claimed as its strongest evidence.

This is not a small thing to say, and it was not said lightly. It required retracting a result the lead author had already published, already celebrated, and already shared widely with genuine belief in its truth. That the correction came from the author's own hand, unprompted by any outside critic, is itself part of what this paper offers: not just a negative result, but a demonstration of what it looks like to test a cherished idea honestly, all the way through, and to report exactly what was found.

Varahamihira's astronomy, and the careful observations of the sages who followed him, remain a remarkable achievement of the ancient world. But on the specific question this study set out to test — can the sky tell us when the ground will break — the sky, on the evidence gathered here, is silent.

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